PESS 1.4 - Thermal Energy REVISION Worksheet

This worksheet is designed to help you build your understanding of thermal energy from the basics through to typical NCEA-style questions, including calculations, graphs, practical situations and written explanations.

It is scaffolded so you can learn the key ideas first, practise applying them, and then attempt more challenging exam-style questions independently.

Learning goals

By the end of this worksheet, you should be able to:

- Distinguish temperature, thermal energy and heat transfer.
- Explain how particles behave when substances are heated or cooled.
- Describe conduction, convection and radiation.
- Explain changes of state using particle theory and latent heat.
- Calculate energy transferred using $E = Pt$ and $E = mc\Delta T$, where appropriate to your course.
- Apply thermal energy concepts to insulation, heating and cooling.
- Show working, use correct units and explain your reasoning in exam-style questions.

Note: Check the exact scope of your school's PESS 1.4 assessment and teacher-provided formula sheet. The questions below are a broad thermal-energy revision resource; not every topic or formula will necessarily be assessed in your particular course.

PLEASE WRITE YOUR ANSWERS IN RED or BLUE

Section A — Essential knowledge and vocabulary

A1. Key terms

Complete the definitions in your own words.

1. Energy:
2. Thermal energy:
3. Temperature:
4. Heat transfer:
5. Specific heat capacity:
6. Latent heat:
7. Power:
8. Conservation of energy:
9. Efficiency:

A2. Temperature versus thermal energy

Answer each question.

1. Explain the difference between temperature and thermal energy.

2. A cup of hot water and a swimming pool of cooler water are compared. Explain why the swimming pool may contain more total thermal energy than the cup.
3. Two identical metal blocks have the same mass and are made of the same material. Block A is at 20°C and Block B is at 80°C. Which block has the greater thermal energy relative to the same reference state? Explain.
4. What happens to the average kinetic energy of particles when the temperature of a substance increases?
5. Does thermal energy always transfer from a colder object to a hotter object naturally? Explain.

A3. Energy forms

The Level 1 standard includes kinetic, gravitational potential, elastic potential, thermal and chemical energy. Electrical energy and energy carried by waves can also be involved in transfers.

Identify the main energy form or forms involved in each situation.

Situation	Energy form(s)
1. Water being heated in a kettle	
2. A moving bicycle	
3. A book held above the floor	
4. A stretched elastic band	
5. Petrol stored in a fuel tank	
6. A hot metal spoon cooling down	
7. Sunlight warming a black car	
8. A battery powering a torch	

Section B — Understanding thermal energy

B1. Heating and cooling

For each situation, describe what happens to the energy of the system.

1. A metal block is heated by an electric heater.
2. A hot cup of tea cools on a table.
3. Water is placed in a freezer.
4. A metal rod is rubbed with sandpaper and becomes warmer.
5. A moving object slows down because of friction.

For each, state:

- Whether the system gains or loses energy.
- The form of energy before and after the change.
- How energy is transferred into, out of or within the system.

B2. How energy is transferred

Energy can be transferred through heating, electricity, waves or doing work.

Match each situation to the main process responsible for the transfer.

Situation	Main transfer process
1. An electric element warms a saucepan	
2. A hand warms a cold metal object by rubbing it	
3. Sunlight warms a person standing outdoors	
4. A battery supplies energy to a circuit	
5. A person lifts a heavy box onto a shelf	
6. Sound from a loudspeaker carries energy to a microphone	

B3. Conduction, convection and radiation

Although these are not the whole scope of PESS 1.4, they are useful contexts for understanding thermal energy transfer.

1. Define conduction.
2. Explain why a metal spoon becomes warm when left in hot soup.
3. Define convection.
4. Explain how convection currents can transfer energy through water heated in a pot.
5. Define radiation as a method of thermal energy transfer.
6. Explain how energy from the Sun reaches Earth.
7. Why does a black surface generally absorb more radiant energy than a shiny, light-coloured surface?
8. Give one practical example of reducing unwanted thermal energy transfer.

B4. True or false?

Write True or False. Correct any false statements.

1. Energy can be created when a heater warms water.
2. A warmer object has a higher temperature than a cooler object.
3. Thermal energy can be transferred electrically to a heating element.
4. An object that cools down transfers energy to its surroundings.
5. All energy transferred into a device becomes useful output energy.
6. Power measures the rate at which energy is transferred.
7. Energy is conserved even when some of it is transferred to the surroundings.
8. A larger object always has a higher temperature than a smaller object.

Section C — Formulae and calculations

These are the main formulae relevant to thermal energy calculations. Use the formula sheet supplied by your teacher if it differs.

Formula reference

1. Energy transferred by power

$$\Delta E = Pt$$

- ΔE = energy transferred (J)
- P = power (W)
- t = time (s)

2. Energy needed to change temperature

$$E = mc\Delta T$$

- E = energy transferred (J)
- m = mass (kg)
- c = specific heat capacity (J kg⁻¹ °C⁻¹)
- ΔT = temperature change (°C)

$$\Delta T = T_{final} - T_{initial}$$

3. Energy needed for a change of state

$$E = mL$$

- L = specific latent heat (J kg⁻¹)

4. Efficiency

\[\text{Efficiency}= \frac{\text{useful energy output}}{\text{total energy input}}\times100\% \]

5. Conservation of energy

$$E_{input} = E_{usefuloutput} + E_{otheroutput}$$

For calculations, assume no energy is lost to the surroundings unless the question states otherwise. Use the specific heat capacity or latent heat value given in each question.

C1. Practising the formula $E = Pt$

Worked example

A 1,000 W heater operates for 60 seconds.

Step 1: Write the formula.

$$E = Pt$$

Step 2: Substitute the values.

$$E = 1000 \times 60$$

Step 3: Calculate.

$$E = 60000J$$

The heater transfers 60,000 J of energy in 60 seconds.

Your turn

1. A 500 W heater operates for 20 seconds. Calculate the energy transferred.
2. A 2,000 W kettle operates for 90 seconds. Calculate the energy transferred.
3. A 1,500 W heater transfers 90,000 J of energy. Calculate the time it operates for.
4. A device transfers 24,000 J in 30 seconds. Calculate its power.
5. Two heaters operate for the same length of time. Heater A is rated at 1,000 W and Heater B at 2,000 W. Compare the energy each transfers in 60 seconds.

C2. Specific heat capacity

For these questions, use the specific heat capacity of water:

$$c = 4200 \text{ J kg}^{-1} \,^{\circ}\text{C}^{-1}$$

Worked example

Calculate the energy required to heat 2 kg of water from 20°C to 30°C.

Step 1: Calculate the temperature change.

$$\Delta T = 30 - 20 = 10^{\circ}C$$

Step 2: Write the formula.

$$E = mc\Delta T$$

Step 3: Substitute the values.

$$E = 2 \times 4200 \times 10$$

Step 4: Calculate.

$$E = 84000J$$

Your turn

1. Calculate the energy needed to heat 1 kg of water from 20°C to 40°C.
2. Calculate the energy needed to heat 0.5 kg of water from 15°C to 55°C.
3. A 2 kg sample of water gains 42,000 J. Calculate its temperature change.
4. A 1.5 kg sample of water is heated from 20°C to 60°C. Calculate the energy transferred.
5. A 3 kg sample of water gains 126,000 J. Calculate the temperature change.
6. Two equal masses of water receive the same energy. Explain why their temperature changes are equal, assuming the same initial conditions and no energy losses.
7. Two samples of water receive the same amount of energy. Sample A has a mass of 1 kg and Sample B has a mass of 2 kg. Compare their temperature changes.

C3. Latent heat and changes of state

During a change of state, energy can be transferred without a change in temperature.

Use:

$$E = mL$$

For the questions below, use the following values:

- Specific latent heat of fusion of ice: $334000\ J\ kg^{-1}$
- Specific latent heat of vaporisation of water: $2260000\ J\ kg^{-1}$

1. Calculate the energy required to melt 0.5 kg of ice at 0°C.
2. Calculate the energy required to melt 2 kg of ice at 0°C.
3. Calculate the energy needed to vaporise 0.2 kg of water at its boiling point.
4. Explain why the temperature remains constant while ice melts at 0°C, even though energy is being transferred into it.
5. Explain why vaporising water requires more energy per kilogram than melting ice.

C4. Mixed calculations

Show your working, include units and give your final answer clearly.

1. A 2,000 W kettle runs for 120 seconds. Calculate the energy transferred.
2. A 0.5 kg metal block has a specific heat capacity of 900 J kg^{-1} $°C^{-1}$ Calculate the energy needed to heat it from 20°C to 80°C.
3. A heater transfers 180,000 J to 1 kg of water. Calculate the temperature rise, assuming no energy losses.
4. A 1,500 W heater operates for 200 seconds. If 240,000 J of energy is usefully transferred to the water, calculate the useful efficiency.
5. An electric kettle transfers 360,000 J of energy into the heating system. The useful energy transferred to the water is 300,000 J. Calculate the energy transferred elsewhere.

6. A 2 kg sample of ice at 0°C melts completely. Calculate the energy required using the supplied latent heat value.

Section D — Conservation of energy and energy bar charts

Energy is conserved: it is transferred between objects or transformed from one form to another, rather than created or destroyed. Some transfers produce energy that is not useful for the intended purpose.

D1. Energy transformations

For each example, identify the initial energy form, the final useful energy form and any other energy transfers.

1. An electric kettle heats water.
2. A person rubs their hands together.
3. A car brakes and slows down.
4. A battery-powered torch is switched on.
5. A ball falls from a height and hits the ground.
6. A student uses an electric heater to warm a classroom.

D2. Energy bar charts

For each scenario, describe what an energy bar chart should show. Use a scale of 0–100 J where appropriate.

Question 1: Heating water

A heater supplies 100 J of energy. The water gains 70 J of thermal energy, while 30 J is transferred to the surroundings.

- Initial energy supplied: _____ J
- Useful energy gained by water: _____ J
- Energy transferred elsewhere: _____ J

Explain how this demonstrates conservation of energy.

Question 2: Braking bicycle

A moving bicycle has 500 J of kinetic energy. It comes to rest. Assume all of its kinetic energy is transferred to the surroundings through friction and other processes.

- Initial kinetic energy: _____ J
- Final kinetic energy: _____ J
- Energy transferred to the surroundings: _____ J

Explain why the bicycle's kinetic energy decreases but energy is still conserved.

Question 3: Lifting a book

A student transfers 20 J of energy to a book by lifting it onto a shelf.

- Which energy store increases?
- What happens to the student's chemical energy store?
- Explain the energy transfer.

D3. Identify the system

In energy questions, always identify the system being studied.

1. A kettle heats water. If the system is the water only, what energy transfer occurs?
2. If the system includes the kettle, water and surroundings, explain why it is important to consider energy transferred to the environment.
3. A ball falls to the ground. If the system includes the ball and Earth, what energy transformation occurs?
4. Explain how choosing a different system boundary can change which energy transfers are counted as inputs or outputs.

Section E — Power and efficiency

E1. Understanding power

Power measures how quickly energy is transferred.

1. Define power and give its unit.
2. Explain the difference between a 1,000 W heater and a 2,000 W heater.
3. Two heaters each transfer 120,000 J of energy. Heater A takes 60 seconds and Heater B takes 120 seconds. Calculate the power of each heater.
4. A 100 W lamp and a 1,000 W heater operate for 10 seconds. Calculate the energy transferred by each.
5. Explain why a higher-power heater can transfer the same amount of energy in less time.

E2. Efficiency calculations

Worked example

A heater receives 500 J of energy and transfers 400 J usefully to the water.

$$\text{Efficiency} = \frac{400}{500} \times 100\% = 80\%$$

The remaining 100 J is transferred elsewhere.

Your turn

1. A device receives 1,000 J and provides 700 J of useful energy. Calculate its efficiency.
2. A kettle receives 200,000 J and transfers 160,000 J usefully to the water. Calculate its efficiency.
3. A motor receives 500 J and transfers 150 J into useful motion. Calculate its efficiency.
4. A heater has an efficiency of 75% and receives 800 J. Calculate the useful energy output.
5. A device receives 600 J and has an efficiency of 40%. Calculate the energy transferred elsewhere.
6. Explain why an electric kettle may be more efficient at heating water than heating a large room with the same water-heating purpose.

Section F — Interpreting data and graphs

F1. Heating water

A student heats 0.5 kg of water using a heater. The water starts at 20°C.

Time (s)	Temperature (°C)
0	20
30	27
60	34
90	41
120	48

Answer the questions.

1. Identify the independent variable.
2. Identify the dependent variable.
3. Describe the relationship between time and temperature.
4. Calculate the total temperature change.
5. Assuming the water's specific heat capacity is 4200 J kg^{-1} $°C^{-1}$, calculate the energy gained by the water.
6. Calculate the average rate of temperature increase in °C/s.
7. Explain why the energy supplied by the heater may be greater than the energy gained by the water.

F2. Comparing two heaters

Two heaters warm identical samples of water under the same conditions.

Time (s)	Heater A temperature (°C)	Heater B temperature (°C)
0	20	20
30	25	30
60	30	40
90	35	50

1. Which heater transfers energy to the water more quickly?
2. Use the data to support your answer.
3. Calculate the temperature increase for each heater after 90 seconds.
4. Assuming the same mass of water, compare the thermal energy gained by the water in each case.
5. State one factor that should be controlled in a fair test comparing heaters.

Section G — Typical NCEA-style written questions

These questions are original practice questions modelled on the concepts and reasoning expected for Level 1 energy assessment. They are not copied examination questions.

Question 1 — Heating a metal block

A 1 kg metal block is heated from 20°C to 70°C. Its specific heat capacity is 900 J kg^{-1} $°C^{-1}$

(a) Calculate the temperature change.

(b) Calculate the energy transferred to the block.

(c) Explain what happens to the block's thermal energy and particle motion as it is heated.

Question 2 — An electric kettle

A kettle rated at 2,000 W operates for 150 seconds.

(a) Calculate the energy transferred by the kettle.

(b) The water receives 250,000 J of useful energy. Calculate the useful efficiency.

(c) Explain why not all the energy supplied by the kettle is transferred usefully to the water.

Question 3 — Melting ice

A student places ice at 0°C into a container and supplies energy until all the ice melts.

(a) Identify the change of state.

(b) Explain why the temperature remains at 0°C while the ice is melting.

(c) Calculate the energy needed to melt 0.3 kg of ice, using $L_f = 334000 Jkg^{-1}$.

Question 4 — Conservation of energy

A ball has 200 J of gravitational potential energy before it falls. Just before reaching the ground, it has 150 J of kinetic energy.

(a) Identify the initial and final energy forms.

(b) Calculate the energy transferred to other forms or the surroundings, assuming the initial and final energy values refer to the same system and all energy changes are accounted for.

(c) Explain why this does not violate the law of conservation of energy.

Question 5 — Comparing two heating systems

Two systems each receive 500,000 J of energy.

- System A transfers 400,000 J usefully to water.

- System B transfers 300,000 J usefully to water.

(a) Calculate the efficiency of System A.

(b) Calculate the efficiency of System B.

(c) Explain which system is more efficient and support your answer with calculations.

(d) Explain what happens to the energy that is not transferred usefully to the water.

Question 6 — Excellence-style integrated explanation

A student claims:

> "If two objects receive the same amount of thermal energy, they will always have the same final temperature."

Discuss this statement. In your answer, consider:

- The mass of each object.
- The material and its specific heat capacity.
- The initial temperatures.
- Energy transferred to the surroundings.

Use a relevant formula to support your explanation.

Section H — Exam technique and self-check

Before submitting your answers, check the following.

My revision checklist

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- ☐ I can distinguish energy, temperature and power.
- ☐ I can identify energy forms and transfer processes.
- ☐ I can use E = Pt correctly.
- ☐ I can use E = mcΔT correctly.
- ☐ I can use E = mL for changes of state.
- ☐ I can calculate efficiency and account for energy transferred elsewhere.
- ☐ I can explain conservation of energy using a system and its initial and final states.
- ☐ I show calculations, units and clear reasoning.
- ☐ I can use data to support an explanation.

Exam reminder: A strong written explanation usually connects the relevant energy form, the transfer or transformation, and the observed change. A strong calculation shows the formula, substitution, answer and unit.

For the official standard and its full scope, refer to the NCEA page for Achievement Standard 92047: PESS 1.4 — Demonstrate understanding of a physical system using energy concepts